

The logo of NISER (National Institute of Science Education Research) is a circular emblem. It features a central design with a DNA helix, a chemical structure, and a lamp. The text "NISER" is prominently displayed in the center. The outer ring of the logo contains the text "राष्ट्रीय विज्ञान शिक्षा एवं अनुसंधान संस्थान" at the top and "विद्ययाऽमृतमश्नुतु" at the bottom.



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About the Standard Operating Procedure (SOP)

Dear User,

The Odisha State Population Based Cancer Registry (OSPBCR) is a commendable initiative taken by the Department of Health and Family Welfare, State Government of Odisha, to ensure accurate reporting of the cancer burden in the state. This initiative is technically supported by Tata Memorial Centre (TMC), Mumbai, and National Institute of Science, Education, and Research (NISER), Jatni, Odisha.

Cancer cases and deaths shall be recorded from several sources across the thirty districts of Odisha, in which you shall play an important role. It is the role of TMC, Mumbai to help OSPBCR attain the best quality possible of data. Therefore, this Standard Operating Procedure (SOP) manual has been prepared to ensure that standardised rules for data collection, case abstraction and coding are followed across all centres.

Data collection and abstraction shall be completed through the web-based application “Setu” developed for OSPBCR, which contains a digitized form of the registry proforma. To enable efficient data reporting, the concise one-page registry proforma contains only those variables that are mostly available during collection process.

In this manual, each variable collected by the registry has been explained in detail. Nevertheless, you are recommended to interact with your senior supervisors and TMC delegates, in case of any technical difficulties. We recommend that before starting the registry operations, staff should read the SOP in detail, with their supervisor.

We look forward to the successful implementation of the OSPBCR, with your support. For any queries you can contact pbcrrodisha@actrec.gov.in / medrec@tmc.gov.in

With warm regards,

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I. Introduction

Cancer registration is an important tool for cancer control. It provides information on the magnitude and patterns of cancer occurrence in a defined area (1). Population-based cancer registries (PBCRs) are the gold standards for providing information on the cancer burden in the defined population. These registries also help in planning cancer control activity as well as gauging the impact of the same. Considering the rising burden of cancer, it is important to understand the true burden of cancer in the population (2).

To systematically document the cancer burden of the whole state, the State Government of Odisha plans to establish a state-wide PBCR for Odisha state. The plan is to represent the cancer burden of all thirty districts of Odisha state, i.e. to cover a population of about 45 million. This step is imperative for improving cancer control and encouraging cancer research in Odisha.

The Odisha State Population Based Cancer Registry (OSPBCR) has been established by the Department of Health and Family Welfare, State Government of Odisha, India, with technical support from Tata Memorial Centre, Mumbai and National Institute of Science, Education, and Research, Jatni, Odisha.

The ability of the cancer registry to deliver accurate information on cancer estimates relies on the quality and completeness of its data. The first step to achieve this involves accurately recording the acquired information of the data items of the registry. This manual is developed to guide the PBCR personnel to abstract the cancer patient data accurately. The guidelines mentioned in this book are based on the standard protocols developed by the International Agency for Research on Cancer (IARC), Lyon, France (3).

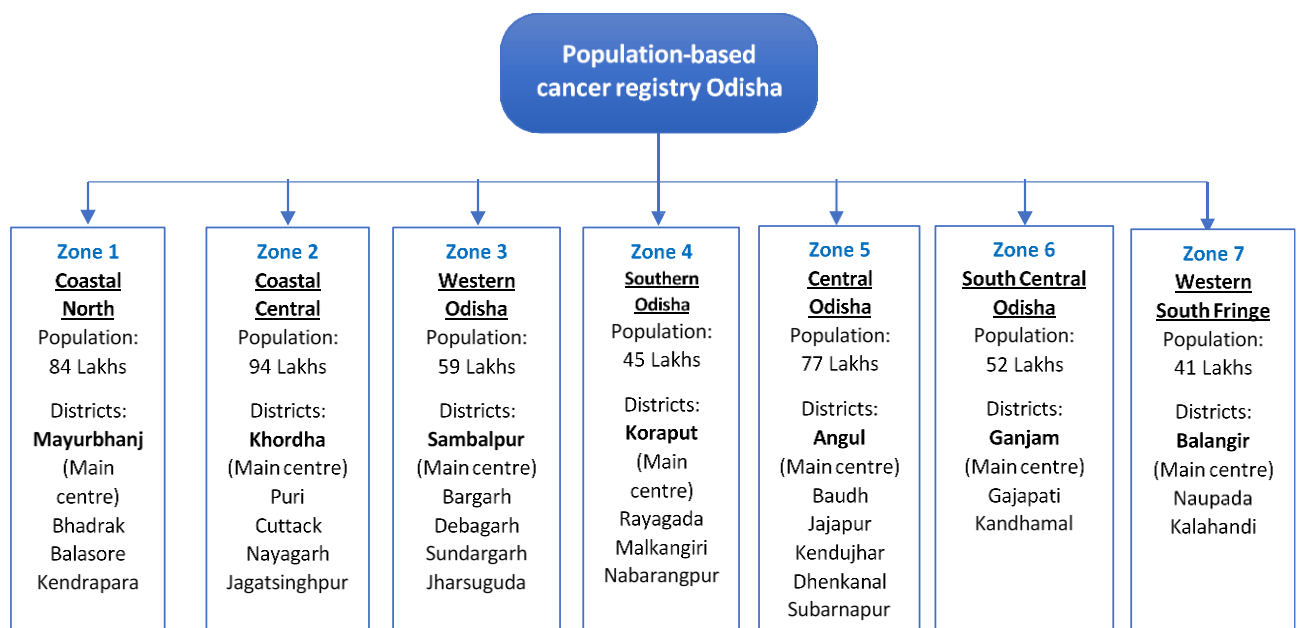
II. Zonal Division of the Odisha state and Work Flow

For practical and implementation purpose of cancer registry, the Odisha state shall be divided in to seven Zones. The map illustrating the zonal division of the Odisha state is mentioned in Figure 1, with constitutive districts mentioned in figure 2.

Figure 1: Zonal division of Odisha state



Figure 2: Districts of each zonal division of Odisha state



Population covered

For the year 2021, the estimated population of the Odisha state is around 4.56 Cr. The total population of males is 2,30,03,001 (50.3%) and for females, it is 2,26,92,999 (49.7%).

Estimated population of the Odisha state (district wise) for the year 2021 mentioned below

Sr. No.	District	2021		
		Male	Female	Total
1	Anugul	7,11,078	6,75,588	13,86,666
2	Balangir	9,00,179	8,95,059	17,95,238
3	Baleshwar	12,85,899	12,40,275	25,26,174
4	Bargarh	8,12,410	8,00,180	16,12,590
5	Baudh	2,40,336	2,39,954	4,80,290
6	Bhadrak	8,24,446	8,15,464	16,39,910
7	Cuttack	14,66,969	13,89,982	28,56,951
8	Debagarh	1,71,589	1,68,639	3,40,228
9	Dhenkanal	6,64,312	6,34,180	12,98,492
10	Gajapati	3,06,765	3,22,365	6,29,130
11	Ganjam	19,29,431	19,12,549	38,41,980
12	Jagatsinghapur	6,26,652	6,11,104	12,37,756
13	Jajapur	10,04,216	9,84,968	19,89,184
14	Jharsuguda	3,21,738	3,09,117	6,30,855
15	Kalahandi	8,53,553	8,63,218	17,16,771
16	Kandhamal	3,90,334	4,07,870	7,98,204
17	Kendrapara	7,78,416	7,89,745	15,68,161
18	Kendujhar	9,83,018	9,78,506	19,61,524
19	Khordha	12,65,674	11,85,400	24,51,074
20	Koraput	7,36,118	7,66,017	15,02,135
21	Malkangiri	3,29,258	3,38,358	6,67,616
22	Mayurbhanj	13,62,271	13,81,036	27,43,307
23	Nabarangapur	6,55,874	6,73,436	13,29,310
24	Nayagarh	5,45,072	5,02,948	10,48,020
25	Nuapada	3,27,456	3,37,104	6,64,560
26	Puri	9,38,441	9,10,853	18,49,294
27	Rayagada	5,11,806	5,42,076	10,53,882
28	Sambalpur	5,71,359	5,62,046	11,33,405
29	Subarnapur	3,37,595	3,26,667	6,64,262
30	Sundargarh	11,50,736	11,28,295	22,79,031
Total		2,30,03,001	2,26,92,999	4,56,96,000

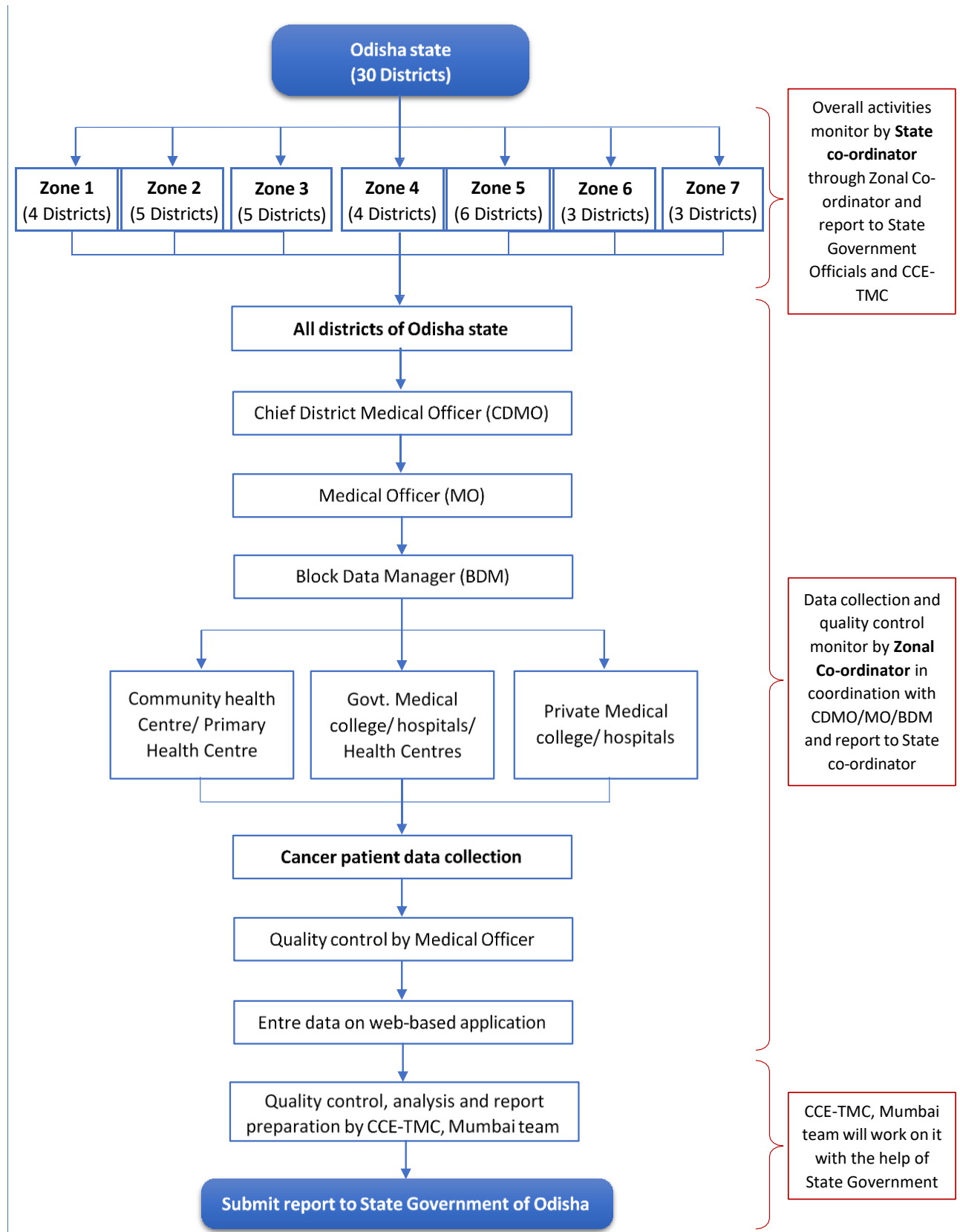
Data collection process

The State will be assigning the responsibility to a Medical Officer/Chemotherapy-in-charge/District Data Manager in each district who will be responsible for collecting data from the block data managers at various community health centres, public and private health centres, NCD cells, diagnostic facilities, and laboratories. They are responsible to share the cancer patient's data within the district through the one-page proforma, mentioned in chapter III. In case of practical difficulties, cases can be discussed with zonal coordinator, state coordinator, and senior member of TMC team. The data shared will be checked by the medical officer after which it will be entered into a web-based software by a data entry operator assigned by the state government. **The data entry will be carried out in the web-based software "Setu," developed for OSPBCR by TMC, Mumbai.** This process will be performed in each district of the Odisha state.

As the districts are grouped in Zones; the Zonal Coordinator shall visit the allocated districts and perform monthly monitoring of the data quality and data entry. The Zonal Coordinator along with the District Hospital staff will visit all the diagnostic and treatment facilities in all the districts of the allocated zone and ensure data collection. The Zonal Coordinator should also coordinate with the Chief District Medical Officer regarding the data collection status of the district. Furthermore, the Zonal coordinator will report the progress to the State Coordinator every month.

The state coordinator shall oversee the cancer registration process across all the zones and periodically visit each zone to update the status of cancer registry data collection, data quality control and monitoring registry related activities. Regular meetings will be conducted in each zone and the state coordinator will provide a progress report to the state government of Odisha, periodically. Additionally, the state coordinator shall conduct quarterly meeting with the state government of Odisha to update the progress regarding cancer registries. The process of data collection and reporting authorities at each level are illustrated in Figure 3.

Figure 3: Process of data collection and reporting authorities at each level



III. Population-based cancer registry: One-page Proforma

POPULATION-BASED CANCER REGISTRY ODISHA STATE, INDIA Project of Ministry of Health and Family Welfare, Government of Odisha Tata Memorial Centre, Mumbai, India													
1. Patient name: _____													
2. AADHAR Number													
3. Age:													
4. Sex:		1. Male			2. Female								
5. Contact number:													
6. Address: _____													
Village: _____				Block: _____				District: _____					
7. Area:		1. Urban			2. Rural			3. Tribal					
8. Hospital name: _____													
9. Hospital case file number: _____													
10. Date of diagnosis:													
11. Basis of diagnosis: (As per ICD-O3 book)													
0. Death Certificate only 1. Clinical 2. Radiology (X-ray, USG, MRI, CT & Endoscopy) 4. Specific Tumor Markers 5. Cytology (Peripheral blood/ Bone marrow) 6. Histology Metastasis 7. Histology of a primary tumor 8. Other/Verbal autopsy													
12. Primary site: _____													
13. Stage:													
1. In situ				2. Localization				3. Direct Extension					
4. Regional lymph node involvement				5. Direct extension with regional node involvement				6. Distant metastasis					
7. Not applicable				9. Unknown/No Information									
14. Histology: _____													
15. Treatment:													
1. Surgery (S)		2. Radio Therapy (RT)		3. Chemotherapy (CT)		4. S+RT		5. S+CT		6. CT+RT			
7. S+RT+CT		8. Hormone therapy (H)		9. S+H		10. RT+H		11. CT+H		12. S+RT+H			
13. S+CT+H		14. RT+CT+H		15. S+RT+CT+H		16. Palliative		17. Other _____		18. No treatment			
19. Unknown													
16. Treatment Status: 1. Complete				2. Incomplete				3. Not applicable				9. Unknown	
17. Vital Status: 1. Alive		2. Dead				9. Unknown							
18. Age at death:													
19. Date of death:													
20. Staff name: _____													
21. Contact number (Staff):													

IV. Variable definitions and examples of coding

1. Patient Name

1. Patient name: _____

This variable requires the name of the patient in the given format:

First name: Name of the patient

Middle Name: Name of father/husband of the patient

Last Name: Family/Lineage name (Surname) of the patient.

The name and surname of the patient is useful for duplicate checking. It is recommended that the full name should be entered as per the government-issued ID card.

If middle name is unavailable, it is acceptable to fill only first name and surname.

For example:

For a patient named Ramesh Prakash Kasale; Ramesh will be the First Name, Prakash is the Middle name and Kasale is the Last Name.

2. AADHAR Number

2. AADHAR Number

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This field contains the 12-digit unique identification AADHAR number of the patient.

What is AADHAR Number?

An Aadhaar number is a 12-digit number that is issued by the Unique Identification Authority of India (UIDAI) to Indian residents as proof of identity and address.

If the AADHAR number is available with the patient, please request to provide. If patient is not willing to provide the AADHAR card number, please mention 999999999999.

3. Age

3. Age:

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The variable refers to the age in years at the incidence date or at the time of cancer diagnosis. In some cases, age may not be known or deliberately recorded inaccurately. It is advised to calculate the age to get the more accurate age information.

If the date of birth is available, then calculate the patient's actual age by using the date of birth and the date of first diagnosis report/clinical note. When the date of birth is unavailable, consider the age mentioned on the first diagnosis report/clinical note.

In *Setu* application, there is a provision of calculation and verification of age at incidence using date of birth and date of incidence. Date of birth has to be entered in *Setu* application.

If the paediatric patient is less than one-year-old, the age should be entered as 0 years.

For example,

If a Baby/infant is 6 months old and is diagnosed with cancer, the age of Baby/infant recorded on the proforma should be 0 years.

4. Sex

4. Sex:	1. Male	2. Female	☐
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This variable requires the gender information of the patient i.e. Male or Female.

5. Contact number

5. Contact number:									
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This variable requires the contact information of the patient. The contact information of the patient and if unavailable, the contact details of the close relative (son, daughter, spouse, brother, sister, father, mother, etc.) should be entered.

If the family doesn't have a mobile phone, then record the mobile number of the ANM/ASHA/Sarpanch/Gram Pradhan/Ward member of that village/ward to trace the patient in the future.

If all of the above-given parameters are unavailable, then mention as 9999999999.

6. Address

6. Address: _____
Village: _____ Block: _____ District: _____

The address is useful for patient identification. It is also essential to identify the registered cases who are residents of the registry area, to calculate incidence rates, and to understand region-wise variation in the incidence of cancer, the address serves to be an important variable. The address is also essential to carry out follow-up for vital status.

This field requires the address of the patient at the time of diagnosis. The address must fulfill the following criteria, first- Address should be from registry area and second- patient must be a resident of the same address for at least a year.

7. Area

7. Area:	1. Urban	2. Rural	3. Tribal	☐
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This variable requires details of the **area of the patient's residence**, in terms of:

Urban: In India, the administrative coverage of the urban area is under Municipal Corporation. According to the Census of India, an area is considered urban if it has a minimum population of 5,000. At least 75% of the male population engaged in non-agricultural occupations. A density of population of at least 400 persons per sq. km.

Rural: In India, the administrative coverage of the rural area is under Gram Panchayat. According to the Census of India, an area is considered rural if it has a population of less than 5,000 people and a population density of less than 400 people per square km, and more than 75% of male working population is engaged in agriculture as an occupation.

Tribal: A tribal area is generally defined as a geographical region with a significant population of indigenous people, characterised by unique cultures, traditions, and a close connection to the land. The list of these tribes are enlisted by the Government of India. Please discuss with district authorities for tribal region.

8. Hospital name

8. Hospital name: _____

Name of the hospital attended by the patient, where data is being entered has to be mentioned here.

9. Hospital case file number

9. Hospital case file number: _____

In this variable, the case file number of the hospital attended by the patient should be mentioned. This shall be useful while back-tracing patient details.

10. Date of Diagnosis (Incidence date)

10. Date of diagnosis:

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The date of diagnosis in a cancer registry refers to the specific date when a healthcare professional confirms and records the diagnosis of cancer in a patient. The date of diagnosis is typically recorded when a biopsy or other diagnostic tests confirm the presence of cancer in an individual. The earliest date is entered usually in the format dd/mm/yyyy.

If the above information is missing, then other dates may be used in an order of declining priorities;

1. Date of first consultation at hospital, clinic, or cancer institution.
2. Date of death (year only), when the cancer is first noted from the death certificate and the follow back attempts have been unsuccessful.
3. Date of death preceding and autopsy, when this is the time at which the cancer is first found and was unsuspected clinically.

Unknown date of diagnosis: There can be situations where the exact date of diagnosis is unknown. For example, if a patient has died or no clinical record is available with the patient/relatives, then get the verbal information of the date of diagnosis if they can provide the date or at least the month and year of diagnosis. If you get the month and year then mention it as 15/mm/yyyy or if you have only year information, then put it as 01/07/yyyy.

Examples:

Scenario 1:

Case 1: Clinically suspected breast cancer on 24-05-2022 (clinical notes available in the registry record), further confirmed with a biopsy report of Ductal Carcinoma was obtained on 15-06-2022. Therefore, the **Date of Diagnosis: 24-05-2022**.

Scenario 2:

Case 2 consulted with the doctor on 17-08-2022 and there was no clinical diagnosis. The patient was suggested endoscopy. The report further confirms with endoscopy report of esophageal malignancy on 20-08-2022. Further, a biopsy was done on 21-08-2022 but no record was found. Therefore, the **Date of Diagnosis: 20-08-2022**.

Scenario 3:

Case 3 consulted with the doctor on 07-03-2022. There was no clinical diagnosis. The patient was suggested oral biopsy. Further confirmation was obtained with a biopsy report of squamous cell carcinoma on 22-03-2022. Therefore, the **Date of Diagnosis: 22-03-2022.**

Scenario 4:

Case 4: Death certificate received from Death Registrar Office with the cause of death cancer and death on 25-07-2022. This case further was confirmed through a house visit. As per a family member the patient was diagnosed with cancer at Hospital 1. The date of diagnosis was not known. The case was verified at Hospital 1 and we received the clinical note dated 18-10-2021 that mentioned stomach cancer. Therefore, the **Date of Diagnosis: 18-10-2021.** (This case is called Death Certificate Notification).

If: The date of diagnosis was not obtained from a family member, and the hospital record was not shared/received at that time the *date of diagnosis and date of death will be similar.*

Scenario 5:

Case 5: The case was found through a village visit where the ANM/ASHA gave information of cancer death. Further staff visited the deceased house and had a verbal autopsy. The staff took the information about signs/symptoms, diagnostic tests, hospital name, treatment received. No data was received from the diagnostic centre/treating hospital. The case was registered on verbal autopsy information and with strong evidence of cancer diagnosis. Verbal information could only provide the year and month of diagnosis as July 2020. Therefore, the **Date of Diagnosis: 15-07-2020.**

11. Basis of diagnosis (as per ICD-O-3)

11. Basis of diagnosis: (As per ICD-O3 book)

0. Death Certificate only 1. Clinical 2. Radiology (X-ray, USG, MRI, CT & Endoscopy) 4. Specific Tumor Markers
5. Cytology (Peripheral blood/ Bone marrow) 6. Histology Metastasis 7. Histology of a primary tumor 8. Other/Verbal autopsy

The "basis of diagnosis" in a cancer registry refers to the specific criteria and evidence used to confirm and document a cancer case for inclusion in the registry. The most valid basis of diagnosis often requires confirmation through histological or cytological examination of tissue samples. In case of unavailability of microscopic diagnostic report, then the basis of diagnosis may be considered based on the availability of the non-microscopic report/clinical note/death certificate.

The codes for the basis of diagnosis are given below,

0. Death certificate only:

If the registry gets data through the only death certificate. Also, there is no other medical record to document.

1. Clinical:

If the disease is diagnosed as cancer and it is mentioned on the clinical notes of the hospital without any diagnostic test reports, then this will be considered clinical. Also, a clinical diagnosis of cancer may be used if the cancer is highly suspected based on clinical examination and no further diagnostic tests.

2. Radiology (X-Ray, USG, MRI, CT scan, endoscopy, mammography, etc.):

All diagnostic techniques, including X-ray, endoscopy, imaging, ultrasound, exploratory surgery (such as laparotomy), mammography, and autopsy, without microscopic diagnosis.

4. Specific tumor markers:

When the diagnosis is made based on a biochemical and/or immunological marker i.e. specific tumor marker test.

5. Cytology (Peripheral blood/bone marrow):

Examination of cells from a primary or secondary site, including fluids aspirated by endoscopy or needle; also includes the microscopic examination of peripheral blood and bone marrow aspirates.

6. Histology Metastasis:

Histological examination of tissue from the metastasis site and autopsy specimens.

7. Histology of primary tumor:

Histologic examination of tissue from the primary tumor, however obtained, including all cutting techniques and bone marrow biopsies; also includes autopsy specimens of the primary tumor. **This is the gold standard of basis for diagnosis.**

8. Other:

If there is no medical record available with the patient/relatives then the registry will take verbal information about the hospital, diagnostic tests, and cancer-directed treatment. Those cases will be considered under this category.

99. Unknown/No information:

When the basis of diagnosis is not available in the records.

Examples:**Scenario 1:**

Case 1: Clinically suspected breast cancer on 24-05-2022 (clinical notes available in the registry record), further confirmed with a biopsy report of Ductal Carcinoma was obtained on 15-06-2022. Therefore, the **Basis of Diagnosis: Histology of primary tumor (Code 7)**.

Scenario 2:

Case 2 consulted with the doctor on 17-08-2022 and there was no clinical diagnosis. The patient was suggested endoscopy. The report further confirms with endoscopy report of esophageal malignancy on 20-08-2022. Further, a biopsy was done on 21-08-2022 but no record was found. Therefore, the **Basis of Diagnosis: Radiology (Code 2)**.

Scenario 3:

Case 3 consulted with the doctor on 07-03-2022. There was no clinical diagnosis. The patient was suggested oral biopsy. Further confirmation was obtained with a biopsy report of squamous cell carcinoma on 22-03-2022. Therefore, the **Basis of Diagnosis: Histology of primary tumor (Code 7)**.

Scenario 4:

Case 4: Death certificate received from Death Registrar Office with the cause of death cancer and death on 25-07-2022. This case was further confirmed through a house visit. As per a family member the patient was diagnosed with cancer at Hospital 1. The date of diagnosis was not known. The case was verified at Hospital 1 and we received the clinical note dated 18-10-2021 that mentioned stomach cancer. Therefore, the **Basis of Diagnosis: Clinical (1)**.

If: The clinical notes/record were not obtained from a family member, and the hospital record was not shared/received at that time the *Basis of diagnosis will be a Death certificate only (0)*.

Scenario 5:

Case 5: The case was found through a village visit where the ANM/ASHA gave information of cancer death. Further staff visited the deceased house and had a verbal autopsy. The staff took the information about signs/symptoms, diagnostic tests, hospital name, treatment received. No data was received from the diagnostic centre/treating hospital. The case was registered on verbal autopsy information and with strong evidence of cancer diagnosis. Verbal information could only provide the year and month of diagnosis as July 2020. Therefore, the **Basis of Diagnosis: Other (Verbal autopsy) (Code 8)**.

12. Primary site

12. Primary site: _____

The primary site of cancer refers to the specific organ or tissue where a cancerous tumor originates. It is the location where the initial cancer cells develop and proliferate.

The detailed topography of a tumor is the most important item to be recorded. It provides the main axis of tabulation of the registry data. The location of the tumor should be clearly and explicitly written in words with as much specific information as possible. The information of this item should be updated with additional information available. ICD-O3 coding system should be used refers to the anatomical location of the primary tumor.

Example 1:

Case No. :	Req No. : FZZSP20019918	Path No. : 015905/CT
Name:	Gender/Age: F / 49 years	
Category: C	DMG: DMG - BREAST SERVICE	
FINAL HISTOPATHOLOGY REPORT		23/04/2020
Nature of Material Received: Right breast lump core biopsy [1 Biopsy]		
Gross Description:		
1) Received multiple cores aggregating to 1.2x0.7cm submitted entirely.		
2) Received multiple cores aggregating to 1x0.6cm submitted entirely.		
Sections:		
1,2) Right breast lump core biopsy		
Microscopic Description:		
Right breast lump core biopsy:		
Invasive breast carcinoma, no special type, grade3.		
Modified Bloom Richardson score: 3+3+2=8.		
Ductal carcinoma in situ (DCIS) is identified, solid type, high nuclear grade with necrosis.		
Results of immunohistochemistry:		
Estrogens Receptor (ER): Negative (Internal control is positive) (Staining repeated)		
Progesterone Receptor (PR): Positive (10% of tumor cells show moderate nuclear staining; Allred score 4/8) (Staining repeated)		
HER2/neu (Ventana): Positive (Score 3+).		
Impression:		
• Right Breast Lump-Core Biopsy : • Invasive breast carcinoma, no special type (NST), grade 3		

Date of diagnosis	23.04.2020 (Needs to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Histology of primary (7)
Primary site	Breast, NOS (C50.9)
Morphology	Invasive breast carcinoma (8500)
Behaviour	Malignant (3)
Grade	Grade 3 (3)

Example 2:

Case No. :	Req No. : FZZSP20016945	Path No. : 013868/CT
Name:	Gender/Age: M / 64 years	
Category: C	DMG: DMG - HEAD & NECK(C)	
FINAL HISTOPATHOLOGY REPORT		18/03/2020
Nature of Material Received: 1 Stained slide[924]		
Impression:		
• Tongue (Right)-Biopsy (1 stained slide) : • Squamous cell carcinoma, moderately differentiated		

Date of diagnosis	18.03.2020 (Needs to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Histology of primary (7)
Primary site	Tongue, NOS (C02.9)
Morphology	Squamous cell carcinoma (8070)
Behaviour	Malignant (3)
Grade	Moderately differentiated (2)

Example 3:

Gender / Age F / 52 Years	Category/Status C/ Out Patient
DMG :DMG GASTROINST.(A)	
Nature of Material	Received one paraffin block labeled as 18-60
FINAL HISTOPATHOLOGY REPORT	
27-01-2018	
GROSS DESCRIPTION	
MICROSCOPIC DESCRIPTION	
IMPRESSION	
Received one paraffin block labeled as 18-60 and designated as, Liver SOL Trucut biopsy: METASTATIC POORLY DIFFERENTIATED CARCINOMA.	

Date of diagnosis	27.01.2018 (Needs to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Histology of metastatic (6)
Primary site	Unknown primary (C80.9)
Morphology	Carcinoma (8010)
Behaviour	Malignant (3)
Grade	Poorly differentiated (3)

Example 4:

Name:	CR.No.	CT No.
Age : 70 Years	Admn.No	Prev.CT No.
Sex : Male	OPD/Ward: OPD	Ref.by
No. of films used: 2		Date: 18/9/17

Clinical history and findings: Right level V lymph node x 1.5mo.

Provisional diagnosis :

Relevant investigation : CECT BOS To T4 (256)

Report:-The study done by taking 5/5mm axial sections from base of the skull to T4 with a pitch of 1.5 following administration of IV contrast.

- Multiple enlarged lymphnodes are seen in right cervical level II, III & V with largest measuring 2.2cm in SAD seen in right level V.
- Few enlarged lymph nodes are seen in visualised mediastinum, largest 13mm in SAD in right paratracheal region. Many of the lymph nodes in prevascular location show calcifications.
- Nasopharynx including the fossa of rosenmuller are normal.
- Oropharynx including base of tongue is normal.
- Larynx including bilateral pyriform sinuses, true and false cords are normal.
- Bilateral submandibular and parotid glands are normal in shape bulk and attenuation.
- B/L lobes of thyroid and isthmus show normal bulk and enhancement.
- Proximal right IJV from C1 to C4 level is attenuated, ?thrombosed. Distal right IJV is dilated. Rest of great vessels of neck and mediastinum are normal.
- B/L visualised lungs show paraseptal emphysematous changes with subpleural bullae. Fibrocalcific changes are seen in b/l upper lobes apical segments.
- Note made of degenerative changes in the cervical spine.

Imp :-Right cervical and mediastinal lymphadenopathy as described metastatic.

Date of diagnosis	18.09.2017 (Needs to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Radiology (2)
Primary site	Unknown primary site (C80.9)
Morphology	Right cervical and mediastinal lymphadenopathy as described metastatic (8000)
Behaviour	Malignant (3)
Grade	Not determined (9)

Example 5:

Case No.:	Requisition No. FZZ/SP/18/002352	Req. Dt. 14-08-2018	Path No. HT/18/1556
Name			
Gender / Age M / 48 Years		Category/Status C/ Out Patient	
DMG : DMG BONE-SOFT TISSUE			
Nature of Material		Right leg biopsy	
FINAL HISTOPATHOLOGY REPORT			20-08-2018
GROSS DESCRIPTION			
Received in formalin labeled with patient name is a specimen of Right leg biopsy containing two tissue bits measuring 0.8x0.5 cm, submitted entirely.			
MICROSCOPIC DESCRIPTION			
IMPRESSION			
Right leg biopsy: SQUAMOUS CELL CARCINOMA WELL DIFFERENTIATED.			

Date of diagnosis	20.08.2018 (Need to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Histology of primary (7)
Primary site	Skin of leg (C44.7)
Morphology	Squamous cell carcinoma (8070)
Behaviour	Malignant (3)
Grade	Well differentiated (1)

Basis of diagnosis: Histology of metastatic (6)

<i>Classification and coding of neoplasms</i> 75 the primary gastric cancer. In such circumstances, the cancer registry should code the primary site, namely stomach, including the morphology of the metastasis, with behaviour /3. If the registry wishes to distinguish between tumours verified by microscopic examination of the primary cancer and those confirmed from histological examination of a metastasis, an additional code specifying the basis of the diagnosis should be used (see Chapter 6, item 17). <u>If, for example, a tumour is reported as being clinically a primary carcinoma of the lung and the diagnosis is supported by microscopic examination of mediastinal lymph nodes showing metastatic squamous-cell cancer, it should be coded as T-162.9 (lung), M-8070/3 (squamous-cell carcinoma). The basis of diagnosis code would in this instance be 6, i.e., histology of metastasis.</u>

Reference: Cancer Registration: Principles and Method (IARC Scientific Publication No. 95) DM Parkin et al.: Chapter No. 7 Classification and Coding of Neoplasms

Primary site and the behaviour code in ICD-O

The amalgamation of information on behaviour (malignant, *in situ*, unknown) and on origin (primary site, metastatic site, unknown) for a given tumour in one behaviour code poses a potential problem for the use of ICD-O by cancer registries. Tumour registries should primarily identify tumours by the topographic site where the tumour originated—in other words, the primary site—and tabulations should be made by primary site. To help identify the primary site in ICD-O, the behaviour code /3 means malignant, primary site. If for some reason the primary site is unknown, but the disease is certainly malignant, the code T-199.9 M—/3 should be used (T-199.9 is the code for unknown primary site.) Sometimes it is clear that there are metastases to, for example, the lungs or liver, but the true site of origin of the tumour cannot be determined. This case should also be coded to T-199.9 M—/3 unknown primary site.

Behaviour: Malignant (3)

code for unknown primary site.) Sometimes it is clear that there are metastases to, for example, the lungs or liver, but the true site of origin of the tumour cannot be determined. This case should also be coded to T-199.9 M—/3 unknown primary site.

Coding of metastases

ICD-O provides for coding the presence of a metastasis in a given organ with a behaviour code /6, but this facility should *not* be used in tumour registries (behaviour code /9—uncertain whether primary or metastatic site—is therefore also redundant). The topography code will refer only to primary site (see above).

The /6 code for behaviour was designed for use by pathologists who receive, for example, tissue from the lung or liver, look under the microscope and recognize a metastasis but do not know where the tumour originated. A pathology laboratory would code this T-162.9 (lung) and M—/6 meaning metastasis from some other organ to lung. Although a tumour registry could follow the same convention, by not doing so, it solves the coding problem posed when the primary site is known but the tumour is histologically diagnosed on the basis of a metastasis. For example, a surgeon may choose to remove a lymph gland close to the stomach rather than taking a biopsy from

Classification and coding of neoplasms

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the primary gastric cancer. In such circumstances, the cancer registry should code the primary site, namely stomach, including the morphology of the metastasis, with behaviour /3. If the registry wishes to distinguish between tumours verified by

Example 6:

Case No.	Requisition No. FZZ/SP/18/000715	Req. Dt. 24-03-2018	Path No. HT/18/0494
Name			
Gender / Age M / 42 Years		Category/Status C/ Out Patient	
DMG :DMG GASTROINST.(A)			
Nature of Material	Cervical lymph node biopsy		
FINAL HISTOPATHOLOGY REPORT			04-04-2018
GROSS DESCRIPTION			
Received multiple grey white tissue bits measuring 1.2x0.8x0.4 cm, submitted entirely.			
MICROSCOPIC DESCRIPTION			
IMPRESSION			
Cervical lymph node biopsy: METASTATIC ADENOCARCINOMA,			

Date of diagnosis	04.04.2018 (Needs to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Histology of metastatic (6)
Primary site	Unknown primary (C80.9)
Morphology	Adenocarcinoma (8140)
Behaviour	Malignant (3)
Grade	Not determined (9)

Example 7:

30-419

सूचक द्वारा मरने हेतु

रजिस्ट्रेशन न (कार्यालय प्रयोग हेतु)

रजिस्ट्रेशन तिथि (कार्यालय प्रयोग हेतु)

1. मृत्यु की तारीख 05-04-2019

2. लिंग - (पु/स्त्री) पु

3. मृतक की उम्र 52 वर्ष

4. मृतक का नाम [redacted]

5. पिता/पति का नाम [redacted]

6. मृत्यु स्थान [redacted]

7. स्थायीपता [redacted]

8. सूचक का नाम [redacted]

9. सूचक का पता [redacted]

10. सूचक का हस्ताक्षर [redacted]

11. सूचक का जन्मदिन [redacted]

12. सूचक का पता [redacted]

13. सूचक का नाम या मृत्यु का कारण [redacted]

14. सूचक का नाम या मृत्यु का कारण [redacted]

15. सूचक का नाम या मृत्यु का कारण [redacted]

16. सूचक का नाम या मृत्यु का कारण [redacted]

17. सूचक का नाम या मृत्यु का कारण [redacted]

18. सूचक का नाम या मृत्यु का कारण [redacted]

19. सूचक का नाम या मृत्यु का कारण [redacted]

20. सूचक का नाम या मृत्यु का कारण [redacted]

Date of diagnosis	05.04.2019 (Needs to be discussed with patient/relatives and confirmed)
Basis of diagnosis	Death certificate only (0)
Primary site	Unknown primary (C80.9)
Morphology	(8000)
Behaviour	Malignant (3)
Grade	Not determined (9)

Example 8:

Remark by field staff:

I went to the patient home, patient was dead on 26-11-2017, suffering from blood cancer, was diagnosed in Oct'17. took treatment from BHU, 3 CT scans done. No record found, all records burn.

Date of diagnosis	01.10.2017
Basis of diagnosis	Verbal autopsy (8)
Primary site	Bone marrow (C42.1)
Morphology	Leukemia, NOS (9800)
Behaviour	Malignant (3)
Grade	Not determined (9)

Example 9:

Remark by field staff:

Sir,

I went to Village Mani Pindra block and meet with pt - husband Mr. XYZ and verbal information pt. Suffering from Ca-Ovary and take the treatment at BHU under by Dr. Madhu Jain and take to CT+medication Now pt is well alive pt. Not provided to any medical record because all paper is keep in another place. so Not any documents provided.

Signature: _____ Date: 08092021

Date of diagnosis	As per the patient/relatives remark
Basis of diagnosis	Verbal information (8)
Primary site	Ovary (C56.9)
Morphology	(8000)
Behaviour	Malignant (3)
Grade	Not determined (9)

Example 10:

RADIOTHERAPY SUMMARY

Name of Patient: _____ Date: 10/01/22
Age/Sex: 63 yrs/ male UHID No: 413134
Consultant: _____ DMRT, DNB RT. No: 647/21

Diagnosis: Ca Prostate with bone mets

Histology: Adenocarcinoma

History of present illness: _____, 63 years old male, initially evaluated under (DM Medical Oncology). He was on regular follow up. Bone Scan done s/o metastasis in multiple vertebra and hip joints. He was then referred to us for Palliative Radiotherapy.

Treatment Plan: - Palliative RT

Intent of the Treatment: Palliative

Plan Dosage: Total dose plan was 20 Gy in 5 fractions.

Date of Start: 03/01/22 Date of Last Radiation: - 10/01/22

Date of diagnosis	Need to be discuss with patient/relatives and confirmed
Basis of diagnosis	Histology of primary (7)
Primary site	Prostate (C61.9)
Morphology	Adenocarcinoma (8140)
Behaviour	Malignant (3)
Grade	Not determined (9)

Multiple primaries

The cancer registry counts tumours, not cancer patients. So, if a person develops more than one tumor in his/her lifespan then this will be counted separately. A primary cancer is one which originates in a primary site or tissue and is thus neither an extension, a recurrence nor a metastasis.

More than one tumor may occur at different sites in the same organ or different organs, with the same or different histology, and the same or other times. The reports should be read properly to differentiate if the multiple tumors are manifestations of a single neoplasm i.e. one primary with metastasis, or if they are different primary tumors.

Multiple primary cases will be counted once in a mortality database. Most probably considering the new cancer site (Topography).

Scenario 1:

Case 1: Patient diagnosed with carcinoma, appendix on 24-05-2015 -----> further confirmed with a biopsy report of carcinoma, descending colon was obtained on 15-06-2017.

Conclusion: According to rules for multiple neoplasms, only one tumor shall be recognized if arising in the same site (Appendix 18.1 and descending colon 18.6) and histologies are the same i.e. carcinoma (8010/3). Therefore, the case will be considered as **SINGLE PRIMARY**.

Scenario 2:

Case 2: Woman diagnosed with papillary serous cystadenocarcinoma of the right ovary on 25.08.2019 -----> Further biopsy of left ovary consists of Immature teratoma on 15.09.2022.

Conclusion: The tumor originated from the ovary (single organ) and has two histologies i.e. papillary serous cystadenocarcinoma (8460/3) and immature teratoma (9080/3) from different groups. Therefore, the case will be considered as **TWO PRIMARIES**.

Scenario 3:

Case 3: Woman complains of breast lump -----> Biopsy shows infiltrating duct carcinoma. -----> After three years diagnosed with skin cancer -----> Pathology report shows basal cell carcinoma.

Conclusion: In this case the tumor developed at two different site/organ (breast and skin) as well as different histologies i.e. infiltrating duct carcinoma (8500/3) and basal cell carcinoma (8090/3). Therefore, the case will be considered as **TWO PRIMARIES**.

13. Stage

13. Stage:	1. In situ 4. Regional lymph node involvement 6. Distant metastasis	2. Localization 7. Not applicable	3. Direct Extension 5. Direct extension with regional node involvement 9. Unknown/No Information	
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This is an important variable which needs to be filled with stage at which the patient was diagnosed, i.e. before cancer directed treatment. Registry staff should discuss all clinical record with supervisor as well as with the consultant and mention the appropriate code. The common definitions for stage at diagnosis are in the following table:

Stage	Meaning	Corresponding TNM
In situ	Non-invasive cancer, behaviour (2)	TisN0M0
Localisation	Invasive tumour that has not spread.	T1N0M0 OR T2N0M0
Direct Extension	Tumour extends to surrounding space, but no metastasis.	T3N0M0 OR T4N0M0
Regional lymph node involvement	Metastasis in regional lymph nodes	T1 OR T2 Any N, except N0 M0
Direct extension with Regional lymph node involvement	Tumour extends to surrounding space and metastasis in regional lymph nodes	T3 OR T4 Any N, except N0 M0
Distant metastasis	Tumour has metastasised to another organ or a distant lymph node.	Any T and any N M1
Not applicable	To be entered for haematological cases: leukaemia, lymphoma and multiple myeloma	
Unknown/No information	To be entered when absolutely no information about the stage of the case before treatment is available.	

Please keep in mind that definitions of staging are different for different cancer sites.

It is recommended to refer following staging manuals

- Piñeros M, Parkin DM, Brierley J, editors (2024). User's Guide to Essential TNM (IARC Technical Publications No. 48). Lyon, France: International Agency for Research on Cancer. Available from: <https://publications.iarc.who.int/647>
- CanStaging+ Link: <https://www.canstaging.org/>

It is recommended that registry staff should check the clinical record thoroughly to abstract proper information about TNM staging. Information about staging is also available in radiology reports. Based on clinical note, mention the TNM staging clinical and/or pathological basis. If TNM staging is not available, then mention as No Information.

14. Histology (Morphology)

14. Histology: _____

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Complete histological diagnosis must be recorded in the cancer registries as stated in the pathology report. Write down the complete impression mentioned on the histology/cytology report. Use the ICD-O3 coding book to code the histology.

- Suppose the histology/cytology report impression is malignant only then mention the morphology code 8001.
- If the basis of diagnosis is radiology, clinical, death certificate only, or verbal information then mention the morphology code 8000 (Please check next point also).
- But morphology code will be accepted only with the following histological codes though the basis of diagnosis is radiology, clinical, death certificate only, or verbal information. For more details, refer the manual CHECK AND CONVERSION PROGRAMS FOR CANCER REGISTRIES (IARC/IACR Tools for Cancer Registries) (4)

Morphology	Code	Morphology	Code
Neoplasm, NOS	8000	Craniopharyngioma	9350
Islet cell tumours, gastrinomas	8150-8154	Glioma	9380
Hepatocarcinoma	8170	Neuroblastoma, NOS	9500
Pituitary tumours	8270-8281	Retinoblastoma, NOS	9510
Melanoma of the eye	8720 & site C69._	Meningioma	9530-9539
Melanoma of skin	8720 & site C44._	Lymphoma, NOS	9590
Sarcoma, NOS	8800	Multiple myeloma	9732
Nephroblastoma, NOS	8960	Waldenstrom macroglobulinemia	9761
Choriocarcinoma, NOS	9100	Leukaemia, NOS	9800
Kaposi sarcoma	9140		

The registry should mention the whole impression available on the microscopic or non-microscopic diagnosis report. In addition, attach the diagnostic report along with the proforma as a record.

Behaviour of the tumor (5th digit code of histology)

If the morphology is coded using ICD-O3, the fifth digit expresses the behaviour of the tumor. The behaviour of neoplasms, also known as tumors, refers to how they grow and acts within the body. Depending on the available information, the behaviour is assigned to the tumors.

In PBCRs, the registry is interested in malignant cases and only the malignant cases are included in the registry while the benign cases are followed up periodically.

The codes of the behaviour are mentioned below:

Code	Description
/0	Benign
/1	Uncertain whether benign or malignant
/2	Carcinoma In-situ
/3	Malignant, primary site
/6	Malignant, metastatic site (This will not be used in the PBCR) (As per the IARC rule)

Grade/Differentiation (6th digit code of histology)

The 6th digit of the morphology code, a single-digit code number designating the grade or differentiation of malignant neoplasms. **Only malignant tumors are graded.**

- If the impression is like “moderately to poorly differentiated adenocarcinoma”, then this will be coded as 8140/33
- “Carcinoma, Anaplastic, NOS” will be coded as 8021/34 similarly “malignant teratoma, anaplastic” (9082/34), “retinoblastoma, differentiated” (9511/31), etc.
- The histology with an ICD-O3 code between 967-969 in the case of lymphomas will be categorized as B-cell as grade (6).

Code	Description
1	Grade I (Low grade) Well-differentiated Differentiated, NOS
2	Grade II (Intermediate grade) Moderately differentiated Moderately well differentiated Intermediate differentiation
3	Grade III (High grade) Poorly differentiated
4	Grade IV Undifferentiated Anaplastic
5	T- Cell
6	B- Cell Pre-B B-Precursor
7	Null cell Non T-non B
8	NK Cell Natural killer cell
9	Grade or differentiation not determined, not stated, or not applicable. For lymphomas and leukemias: Cell type not determined, not stated, or not applicable

Examples for histology, behaviour, and grade have been enlisted in the primary site section.

15. Treatment:

15. Treatment:						
1. Surgery (S)	2. Radio Therapy (RT)	3. Chemotherapy (CT)	4. S+RT	5. S+CT	6. CT+RT	
7. S+RT+CT	8. Hormone therapy (H)	9. S+H	10. RT+H	11. CT+H	12. S+RT+H	13. S+CT+H
14. RT+CT+H	15. S+RT+CT+H	16. Palliative	17. Other	18. No treatment	99. Unknown	

The type of cancer directed treatment will be entered in this variable. The codes are mentioned as per the different combinations.

- If the patient has been suggested treatment like surgery, chemotherapy, or radiotherapy and he/she has not undergone surgery or has not received a single dose of CT or RT; then this will be considered as **NO TREATMENT**.
- If the treatment information is not available, then record it as **UNKNOWN**.

16. Treatment status

16. Treatment Status: 1. Complete 2. Incomplete 3. Not applicable 9. Unknown	
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The treatment completion status is mentioned in the field; the codes are mentioned below.

Code	Description
1 Complete	If the patient has received the complete advice treatment
3 Incomplete	If the patient has received partial treatment and left further treatment due to any kind of reason including death.
8 Not applicable (NA)	If the treatment type is PALLIATIVE, OTHER (Non-cancer directed treatment), NO TREATMENT or UNKNOWN
99 Unknown/ No information	If the treatment status information is not available, then record it as UNKNOWN/ NO INFORMATION

Scenario 1:

Case 1: Patient diagnosed with cancer advised surgery and chemotherapy (6 cycles) -----> patient underwent surgery as well as completed all cycles of chemotherapy. The treatment status will be recorded as **COMPLETE**.

Scenario 2:

Case 2: Patient diagnosed with cancer advised surgery and chemotherapy (6 cycles) -----> patient underwent surgery and received only 4 cycles of chemotherapy -----> due to poor health condition, patient left remaining treatment. The treatment status will be recorded as **INCOMPLETE**.

17. Vital status

17. Vital Status: 1. Alive 2. Dead 9. Unknown	
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The vital status obtained during the follow-up is mentioned with the assigned codes as mentioned below.

Code	Description
1 Alive	If the patient is alive
2 Dead	If the patient is dead, then mention the date of death also
99 Unknown/ No information	If the patient lost to follow-up/trace and vital status information is not available, then record it as UNKNOWN/ NO INFORMATION

The date at which the patient was last known to be alive may be known from a follow-up visit to the hospital. This can be obtained by contacting the patient, or the relative. This date is important for computing the survival rates. Date of last follow up should be added as a remark in Setu.

If the patient dies, the date of last contact preferably, should be made identical to the date of death.

Every six or twelve months, the registry should collect the cancer patient's follow-up information. Both the software and the registry proforma should have the follow-up date updated.

18. Age at death

18. Age at death:

--	--

The age of the patient when declared dead has to be mentioned in this variable.

19. Date of death

19. Date of death:

--	--	--	--	--	--	--	--

The date when the patient was declared dead has to be mentioned in this variable.

20. Staff name

20. Staff name: _____

Name of person completing form should be filled in this variable. Registry staff should check that all information filled in the proforma as well as make sure the available reports are attached to the proforma.

21. Contact Number (Staff)

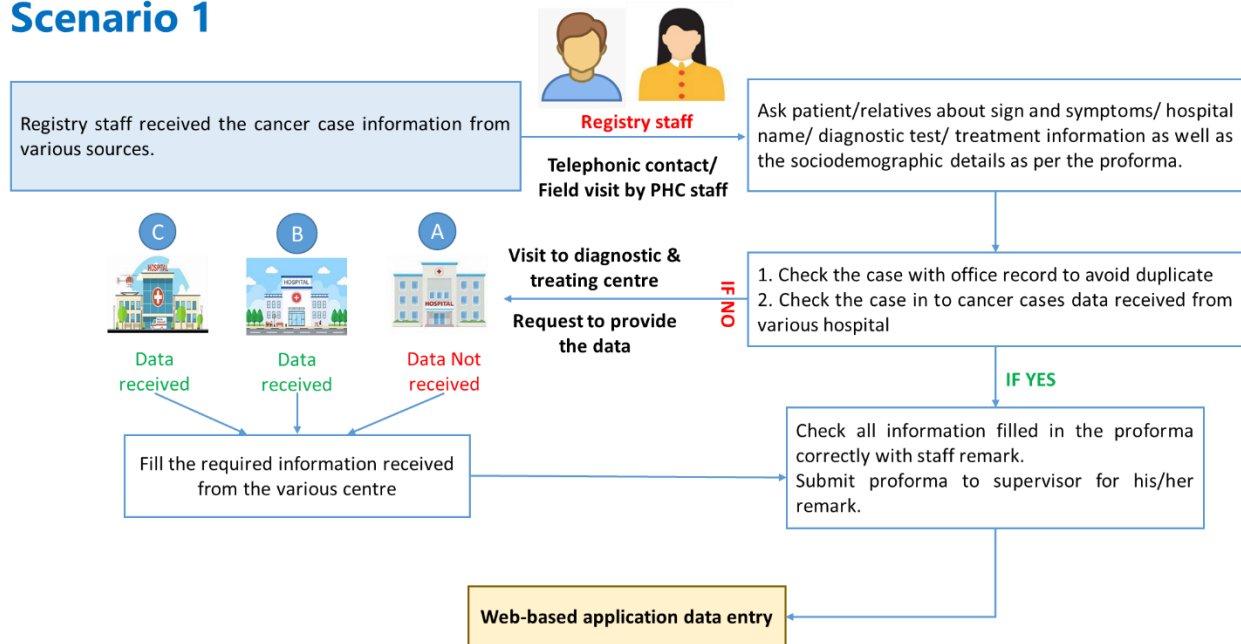
21. Contact number (Staff):

--	--	--	--	--	--	--	--	--	--

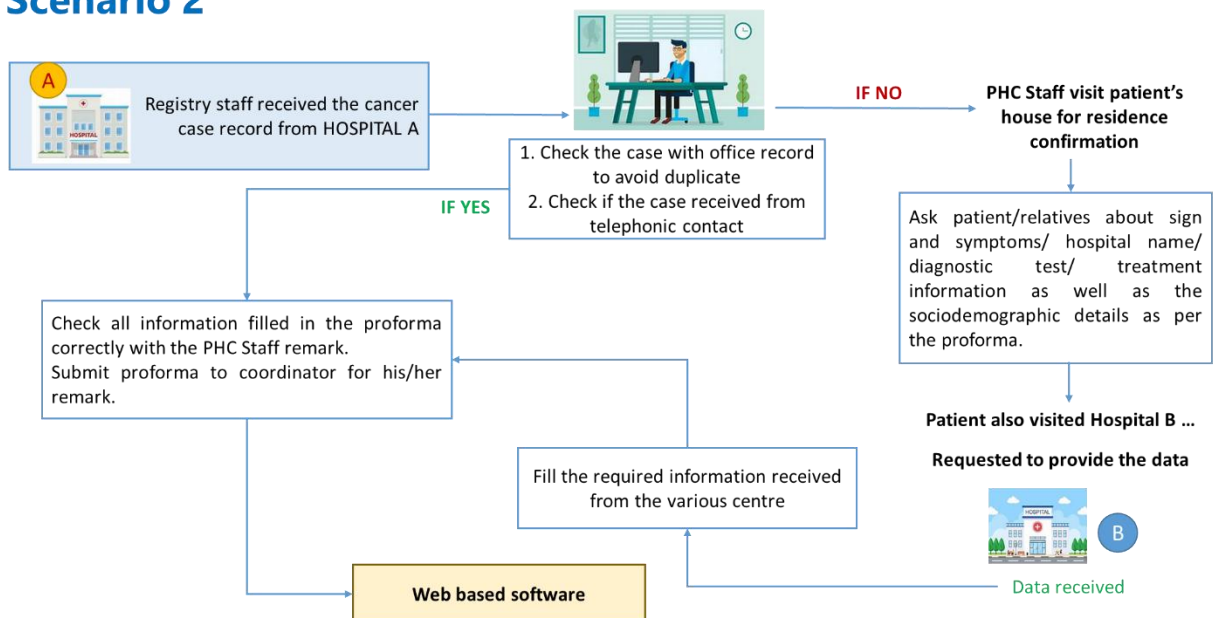
This variable is necessary for verification of details by senior staff.

V. Case Confirmation Procedure

Scenario 1



Scenario 2



VI. References

1. Cancer Registration Principles and Methods. IARC scientific publications No. 95. International Agency for Research on Cancer, Lyon, France (1991). Available online: <https://publications.iarc.who.int/Book-And-Report-Series/Iarc-Scientific-Publications/Cancer-Registration-Principles-And-Methods-1991>
2. Bray F, Znaor A, Cueva P, Korir A, Swaminathan R, Ullrich A, Wang SA, Parkin DM. Planning and Developing Population-Based Cancer Registration in Low- or Middle-Income Settings. Lyon (FR): International Agency for Research on Cancer; 2014. PMID: 33502836. Available online: <https://publications.iarc.who.int/publications/media/download/6625/3132fceb4121af1f3395ad2ddb94d3b4446c2.pdf>
3. MacLennan R. Chapter 6: Items of patient information which may be collected by the registries. Cancer Registration: Principles and Methods. IARC Scientific Publication No. 95. International Agency for Research on Cancer, Lyon, France (1991).
4. Ferlay J, Burkhard C, Whelan S, Parkin DM. CHECK AND CONVERSION PROGRAMS FOR CANCER REGISTRIES (IARC/IACR Tools for Cancer Registries). IARC Technical Report No. 42, Lyon (2005). Available online: http://www.iacr.com.fr/index.php?option=com_content&view=article&id=72:iaccrg_tools&catid=68&Itemid=445

Population-Based Cancer Registry

Odisha State, India

